Milestone 1 – Project Definition

Contents

1. Problem Statement 2
2. Data Sources 2
   2.1. Operational Data Lake 2
   2.2. Access Network Planning Division Data 3
3. Intended Methods 3
   3.1. Key Methods 3
      3.1.1. Traffic Forecasting 3
      3.1.2. Power Consumption Prediction 3
      3.1.3. Site Segmentation 3
      3.1.4. Anomaly Detection 3
   3.2. Key deliverables 3
      3.2.1. Problem Types 3
      3.2.2. Data Types 3
      3.2.3. Technique Categories 3
   3.3. Evaluation Strategy 4
      3.3.1. For regression and forecasting: 4
      3.3.2. For anomaly detection: 4
      3.3.3. Business-focused evaluation will consider: 4
4. Expected Outputs 4
5. Key Risks and Assumptions 4
   5.1. Risks 4
   5.2. Assumptions 4
6. Summary 4

Milestone 1 – Project Definition

Project Title

Data-Driven Power System Upgrade Planning for Cellular Network Sites Using Traffic and Hardware Configuration

1. Problem Statement

Telecommunication operators continuously upgrade radio access network (RAN) infrastructure to improve coverage, throughput, and service quality. A leading telecommunication operator is currently planning a nationwide program of hardware upgrades at selected cellular sites, including the installation of new Radio Remote Units (RRUs), Active Antenna Units (AAUs), baseband equipment, and associated indoor equipment. These new components typically increase site-level electricity demand due to higher transmitting power, processing requirements, and cooling loads.

To safely support these upgrades, existing site power systems such as grid supply configuration, rectifiers, and battery banks, etc. must often be upgraded in parallel. Incorrectly sized power systems create two major risks:

- under-provisioning can lead to service outages and accelerated battery degradation
- over-provisioning results in unnecessary capital expenditure and inefficient utilization of electrical infrastructure.

Currently, such power upgrade decisions rely heavily on engineering rules-of-thumb, manual calculations, and limited historical comparisons across sites. However, the leading telecommunication operator in this context maintains a large internal data lake containing high-resolution traffic telemetry, historical power consumption, and detailed site configuration data. This project proposes exploiting these data assets to create a data-driven decision-support system for power upgrade planning.

The primary objective of this capstone is to develop an end-to-end analytics and machine-learning pipeline that can

- predict incremental power requirements for sites undergoing hardware upgrades
- identify anomalous or inefficient sites relative to similar peers
- prioritize sites for power system upgrades based on quantified risk

The intended users of this system are engineers within the Passive Infrastructure Planning Division, who are responsible for approving hardware deployments and associated electrical upgrades.

2. Data Sources

2.1. Operational Data Lake

The following data will be extracted from the leading telecommunication operator's internal systems:

- Site-level network configuration:
    - RRU & AAU types and quantities
    - cabinet types, baseband unit (BBU) models
    - other electrical loads
- Historical electrical power consumption per site
- Telecommunication traffic recorded every 15 minutes over the past six months, including voice and packet-data volumes
- Existing power supply arrangements:
    - single-phase or three-phase grid connections
    - rectifier capacities
    - battery bank counts and amp-hour ratings

2.2. Access Network Planning Division Data

Additional planning datasets will include:

- Lists of sites scheduled for hardware upgrades
- Proposed new equipment configurations
- Forecast traffic growth or post-upgrade traffic projections

3. Intended Methods

The project will explicitly compare baseline engineering or statistical methods with more advanced machine-learning approaches.

3.1. Key Methods

3.1.1. Traffic Forecasting

To estimate future site load:

- **Baseline**: ARIMA or Prophet models using historical traffic
- **Advanced**: Gradient-boosted regression or neural temporal models incorporating calendar effects and site metadata

3.1.2. Power Consumption Prediction

To model the relationship between traffic, hardware, and energy use:

- **Baseline**: Linear regression or rule-based power calculations derived from vendor specifications
- **Advanced**: Random Forest

3.1.3. Site Segmentation

Sites will be grouped into homogeneous clusters using K-means or hierarchical clustering based on traffic profiles, hardware mix, and geography. Anomalies will be assessed relative to peers within the same cluster.

3.1.4. Anomaly Detection

- **Baseline:** Statistical thresholding using z-scores or inter-quartile ranges
- **Advanced:** Isolation Forest or autoencoder-based detection methods

3.2. Key deliverables

3.2.1. Problem Types

- Regression and forecasting (prediction of site-level power consumption)
- Anomaly detection (identification of inefficient or abnormal sites)
- Comparative analysis across clusters of similar sites

3.2.2. Data Types

- Multivariate time-series data (minute-level voice and data traffic)
- Tabular data (hardware inventories, power system characteristics)
- Aggregated site-level profiles

3.2.3. Technique Categories

- Machine learning
- Time-series forecasting
- Anomaly detection
- Descriptive and diagnostic analytics

3.3. Evaluation Strategy

Model performance will be assessed using both technical and business-oriented metrics.

3.3.1. For regression and forecasting:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Mean Absolute Percentage Error (MAPE)

3.3.2. For anomaly detection:

- Precision@k based on expert review
- cluster-relative deviation scores
- case-study analyses of flagged sites

3.3.3. Business-focused evaluation will consider:

- proportion of sites predicted to exceed existing power capacity
- potential avoided over-provisioning
- prioritization accuracy for urgent upgrades

4. Expected Outputs

- predicted incremental power demand for each upgrade candidate site
- overload probability scores
- ranked lists of sites by urgency of power upgrades
- clusters of comparable sites
- anomaly flags for inefficient configurations
- engineering recommendation tables

5. Key Risks and Assumptions

5.1. Risks

- missing or noisy telemetry records
- inaccuracies in inventory databases
- uncertainty in traffic growth forecasts
- seasonal or event-driven demand spikes
- confidentiality constraints limiting public data release

5.2. Assumptions

- traffic volume is strongly correlated with electrical load
- vendor specifications reflect real-world power usage
- no major topology or backhaul changes occur during the study period
- equipment efficiency characteristics remain stable

6. Summary

This Project Definition presents an initial design for an end-to-end data science pipeline to support power system upgrade planning at cellular network sites, including forecasting, anomaly detection, and decision-support analytics. The problem framing, data sources, intended methods, evaluation strategy, and expected outputs have been proposed based on preliminary understanding of the operational context and anticipated data availability.

As this document has been prepared prior to detailed exploration analysis of the datasets, the above-mentioned elements such as intended methods, key deliverables, evaluation strategy, expected outputs, etc. are subjected to variations. Any such variations will remain within the defined project scope and will be clearly justified in later milestones.